Yizhang ZHU

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EDUCATION

The Hong Kong University of Science and Technology (Guangzhou) **2025.09 - Present**
Ph.D. Student in Data Science and Analytics
Supervisor: Prof. Yuyu LUO

The Hong Kong University of Science and Technology (Guangzhou) **2023.09 - 2025.07**
M.Phil. in Data Science and Analytics
Supervisor: Prof. Yuyu LUO

Chongqing University **2019.09 - 2023.06**
B.Eng. in Computer Science and Technology

EXPERIENCE

Tsinghua University **2025.06 - 2025.09**
Visiting Student in Database Group, Department of Computer Science and Technology
Supervisor: Prof. Guoliang LI

National University of Singapore Chongqing Research Institute **2022.09 - 2023.05**
Visiting Student in Computer Engineering Joint Program
Supervisor: Prof. Yung Chii LIANG

SELECTED PUBLICATIONS

Fist-authored Publications

[1] A Survey of Data Agents: Emerging Paradigm or Overstated Hype? Link
Yizhang ZHU, Liangwei WANG, Chenyu YANG, *et al.*, Guoliang LI, Yuyu LUO
Preprint (Survey)

[2] EllieSQL: Cost-Efficient Text-to-SQL with Complexity-Aware Routing Link
Yizhang ZHU, Runzhi JIANG, Boyan LI, Nan TANG, Yuyu LUO
Conference on Language Modeling (COLM 2025)

[3] Are Large Language Models Good Statisticians? Link
Yizhang ZHU, Shiyin DU, Boyan LI, Yuyu LUO, Nan TANG
Advances in Neural Information Processing Systems (NeurIPS 2024)

Co-authored Publications (View all in Google Scholar)

[4] LEAD: Iterative Data Selection for Efficient LLM Instruction Tuning Link
Xiaotian LIN, Yanlin QI, Yizhang ZHU, Themis Palpanas, Chengliang CHAI, Nan TANG, Yuyu LUO
International Conference on Very Large Data Bases (VLDB 2026)

[5] RAMer: Reconstruction-based Adversarial Model for Multi-party Multi-modal Multi-label Emotion Recognition Link
Xudong YANG, Yizhang ZHU, Nan TANG, Yuyu LUO
International Joint Conference on Artificial Intelligence (IJCAI 2025)

[6] DeepEye: A Steerable Self-driving Data Agent System Link
Boyan Li, Yiran Peng, Yupeng Xie, Sirong Lu, Yizhang ZHU, Xing Mu, Xinyu Liu, Yuyu Luo
Under Review

[7] Boosting Text-to-Chart Retrieval through Training with Synthesized Semantic Insights Link
Yifan WU, Lutao YAN, Yizhang ZHU, Yinan MEI, Jiannan WANG, Nan TANG, Yuyu LUO
Under Review

[8] AskChart: Universal Chart Understanding through Textual Enhancement Link
Xudong YANG, Yifan WU*, Yizhang ZHU*, Nan TANG, Yuyu LUO
Under Review

PROJECTS

DeepEye - Core Developer Link

A Steerable Self-driving Data Agent System (AI Agent 2025 Best Open-Source Project)

- A workflow-centric data agent system that orchestrates complex analytics as transparent DAGs, enabling joint analysis across heterogeneous sources such as databases, files, and knowledge bases.
- Features a database-inspired Workflow Engine (Compiler, Validator, Optimizer, Executor) that ensures structural correctness and efficiency through runtime topological optimization and parallel execution.
- Through hierarchical reasoning with context isolation and a Unified Node Protocol, synthesizes structured and unstructured data insights into Data Videos, interactive Dashboards, and Analytical Reports, moving beyond text-only analysis.

GNN4SL - Project Leader Link

LLM-Enhanced Semantic-Aware Graph Learning for Schema Linking in NL2SQL

- A semantic-aware schema linking approach that recasts NL2SQL schema linking as a graph link-prediction task, aligning user queries with database schema elements via a unified graph formulation.
- Combines LLM-derived semantic embeddings with GNN-based structural learning (GCN/GAT/RGAT) to capture both meaning-level similarity and complex relational dependencies among schema components.
- Builds graph-structured datasets and evaluation pipelines on Spider and BIRD to study effectiveness and efficiency under realistic multi-table schemas and challenging linking patterns.

FUNDINGS AND AWARDS

AI Agent 2025 Competition - *Best Open-source Project Award*	**2025.10**
HKUST(GZ) Ph.D. Student Fellowship	**2025.09**
Data Science and Analytics Thrust Volunteer Grant	**2025.05**
Greater Bay Area CS Academic Poster Competition - *Most Popular Poster Award*	**2025.03**
HKUST(GZ) Red Bird M.Phil. Studentship	**2023.09**
Excellent Graduates of Chongqing University	**2023.06**
Comprehensive Scholarship of Chongqing University	**2022.09**
National Mathematical Contest of Modeling - *First Prize in Chongqing*	**2021.10**
National Undergraduate Innovation and Entrepreneurship Project - *¥50,000 funding*	**2021.05**

ACADEMIC ACTIVITIES

Talks & Seminars:

- Huawei Squirrel Webinar (Invited) - *A Survey of Data Agents: Emerging Paradigm or Overstated Hype?*
- Huawei Paper Discussion Session: Insights into VLDB 2025 Industry Papers
- HKUST(GZ) RBM Project Roadshow - *Towards Automated Statistical Analysis: The Role of LLMs*

Teaching Assistant:

- DSAA6100: Data-centric Artificial Intelligence Practical Lab (Master's Level), HKUST(GZ)

Academic Services: NeurIPS 2025 Reviewer

SKILLS

English Proficiency: IELTS: 7.0 (Listening: 7.5, Reading: 8.5, Speaking: 6, Writing: 6.5)

Professional Skills:

- Programming: Python, SQL, Java, C/C++, Verilog, JavaScript
- AI/Data Science: PyTorch, PEFT, vLLM, TRL, PyG, LangChain; Hadoop, Spark
- Development: Git; FastAPI; Vue, Streamlit; JMeter
- Computer Architecture/Hardware: FPGA, Vivado; Arduino

Academic Writing Skills: LaTeX, Microsoft Visio, OmniGraffle, Figma